



TIME-LLM: TIME SERIES FORECASTING BY REPROGRAMMING LARGE LANGUAGE MODELS

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ABSTRACT

Time series forecasting holds significant importance in many real-world dynamic systems and has been extensively studied. Unlike natural language process (NLP) and computer vision (CV), where a single large model can tackle multiple tasks, models for time series forecasting are often specialized, necessitating distinct designs for different tasks and applications. While pre-trained foundation models have made impressive strides in NLP and CV, their development in time series domains has been constrained by data sparsity. Recent studies have revealed that large language models (LLMs) possess robust pattern recognition and reasoning abilities over complex sequences of tokens. However, the challenge remains in effectively aligning the modalities of time series data and natural language to leverage these capabilities. In this work, we present TIME-LLM, a reprogramming framework to repurpose LLMs for general time series forecasting with the backbone language models kept intact. We begin by reprogramming the input time series with text prototypes before feeding it into the frozen LLM to align the two modalities. To augment the LLM’s ability to reason with time series data, we propose Prompt-as-Prefix (PaP), which enriches the input context and directs the transformation of reprogrammed input patches. The transformed time series patches from the LLM are finally projected to obtain the forecasts. Our comprehensive evaluations demonstrate that TIME-LLM is a powerful time series learner that outperforms state-of-the-art, specialized forecasting models. Moreover, TIME-LLM excels in both few-shot and zero-shot learning scenarios. The code is made available at <https://github.com/KimMeen/Time-LLM>

1 INTRODUCTION

Time series forecasting is a critical capability across many real-world dynamic systems (Jin et al., 2023a), with applications ranging from demand planning (Leonard, 2001) and inventory optimization (Li et al., 2022) to energy load forecasting (Liu et al., 2023a) and climate modeling (Schneider & Dickinson, 1974). Each time series forecasting task typically requires extensive domain expertise and task-specific model designs. This stands in stark contrast to foundation language models like GPT-3 (Brown et al., 2020), GPT-4 (OpenAI, 2023), Llama (Touvron et al., 2023), *inter alia*, which can perform well on a diverse range of NLP tasks in a few-shot or even zero-shot setting.

Pre-trained foundation models, such as large language models (LLMs), have driven rapid progress in computer vision (CV) and natural language processing (NLP). While time series modeling has not benefited from the same significant breakthroughs, LLMs’ impressive capabilities have inspired their application to time series forecasting (Jin et al., 2023b). Several desiderata exist for leveraging LLMs to advance forecasting techniques: **Generalizability**. LLMs have demonstrated a remarkable capability for few-shot and zero-shot transfer learning (Brown et al., 2020). This suggests their

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potential for generalizable forecasting across domains without requiring per-task retraining from scratch. In contrast, current forecasting methods are often rigidly specialized by domain. **Data efficiency.** By leveraging pre-trained knowledge, LLMs have shown the ability to perform new tasks with only a few examples. This data efficiency could enable forecasting for settings where historical data is limited. In contrast, current methods typically require abundant in-domain data. **Reasoning.** LLMs exhibit sophisticated reasoning and pattern recognition capabilities (Mirchandani et al., 2023; Wang et al., 2023; Chu et al., 2023). Harnessing these skills could allow making highly precise forecasts by leveraging learned higher-level concepts. Existing non-LLM methods are largely statistical without much innate reasoning. **Multimodal knowledge.** As LLM architectures and training techniques improve, they gain more diverse knowledge across modalities like vision, speech, and text (Ma et al., 2023). Tapping into this knowledge could enable synergistic forecasting that fuses different data types. Conventional tools lack ways to jointly leverage multiple knowledge bases. **Easy optimization.** LLMs are trained once on massive computing and then can be applied to forecasting tasks without learning from scratch. Optimizing existing forecasting models often requires significant architecture search and hyperparameter tuning (Zhou et al., 2023b). In summary, LLMs offer a promising path to make time series forecasting more general, efficient, synergistic, and accessible compared to current specialized modeling paradigms. Thus, adapting these powerful models for time series data can unlock significant untapped potential.

The realization of the above benefits hinges on the effective alignment of the modalities of time series data and natural language. However, this is a challenging task as LLMs operate on discrete tokens, while time series data is inherently continuous. Furthermore, the knowledge and reasoning capabilities to interpret time series patterns are not naturally present within LLMs’ pre-training. Therefore, it remains an open challenge to unlock the knowledge within LLMs in activating their ability for general time series forecasting in a way that is accurate, data-efficient, and task-agnostic.

In this work, we propose TIME-LLM, a reprogramming framework to adapt large language models for time series forecasting while keeping the backbone model intact. The core idea is to *reprogram* the input time series into text prototype representations that are more naturally suited to language models’ capabilities. To further augment the model’s reasoning about time series concepts, we introduce *Prompt-as-Prefix* (PaP), a novel idea in enriching the input time series with additional context and providing task instructions in the modality of natural language. This provides declarative guidance about desired transformations to apply to the reprogrammed input. The output of the language model is then projected to generate time series forecasts. Our comprehensive evaluation demonstrates that large language models can act as effective few-shot and zero-shot time series learners when adopted through this reprogramming approach, outperforming specialized forecasting models. By leveraging LLMs’ reasoning capability while keeping the models intact, our work points the way toward multimodal foundation models that can excel on both language and sequential data tasks. Our proposed reprogramming framework offers an extensible paradigm for imbuing large models with new capabilities beyond their original pre-training. Our main contributions in this work can be summarized as follows:

- We introduce a novel concept of *reprogramming* large language models for time series forecasting without altering the pre-trained backbone model. In doing so, we show that forecasting can be cast as yet another “language” task that can be effectively tackled by an off-the-shelf LLM.
- We propose a new framework, TIME-LLM, which encompasses reprogramming the input time series into text prototype representations that are more natural for the LLM, and augmenting the input context with declarative prompts (e.g., domain expert knowledge and task instructions) to guide LLM reasoning. Our technique points towards multimodal foundation models excelling in both language and time series.
- TIME-LLM consistently exceeds state-of-the-art performance in mainstream forecasting tasks, especially in few-shot and zero-shot scenarios. Moreover, this superior performance is achieved while maintaining excellent model reprogramming efficiency. Thus, our research is a concrete step in unleashing LLMs’ untapped potential for time series and perhaps other sequential data.

2 RELATED WORK

Task-specific Learning. Most time series forecasting models are crafted for specific tasks and domains (e.g., traffic prediction), and trained end-to-end on small-scale data. An illustration is in

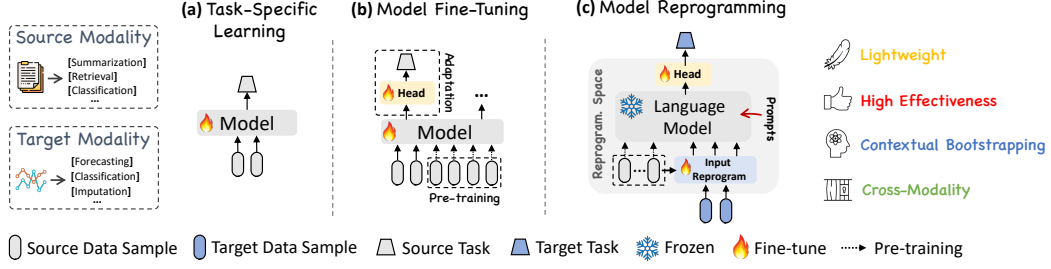


Figure 1: Schematic illustration of reprogramming large language models (LLMs) in comparison of (a) task-specific learning and (b) model fine-tuning. Our proposal investigates and demonstrates (c) how to effectively reprogram open-sourced LLMs as powerful time series learners where well-developed time series pre-trained models are not readily available.

Fig. 1(a). For example, ARIMA models are designed for univariate time series forecasting (Box et al., 2015), LSTM networks are tailored for sequence modeling (Hochreiter & Schmidhuber, 1997), and temporal convolutional networks (Bai et al., 2018) and transformers (Wen et al., 2023) are developed for handling longer temporal dependencies. While achieving good performance on narrow tasks, these models lack versatility and generalizability to diverse time series data.

In-modality Adaptation. Relevant research in CV and NLP has demonstrated the effectiveness of pre-trained models that can be fine-tuned for various downstream tasks without the need for costly training from scratch (Devlin et al., 2018; Brown et al., 2020; Touvron et al., 2023). Inspired by these successes, recent studies have focused on the development of time series pre-trained models (TSPTMs). The first step among them involves time series pre-training using different strategies like supervised (Fawaz et al., 2018) or self-supervised learning (Zhang et al., 2022b; Deldari et al., 2022; Zhang et al., 2023). This allows the model to learn representing various input time series. Once pre-trained, it can be fine-tuned on similar domains to learn how to perform specific tasks (Tang et al., 2022). An example is in Fig. 1(b). The development of TSPTMs leverages the success of pre-training and fine-tuning in NLP and CV but remains limited on smaller scales due to data sparsity.

Cross-modality Adaptation. Building on in-modality adaptation, recent work has further explored transferring knowledge from powerful pre-trained foundations models in NLP and CV to time series modeling, through techniques such as multimodal fine-tuning (Yin et al., 2023) and model reprogramming (Chen, 2022). Our approach aligns with this category; however, there is limited pertinent research available on time series. An example is Voice2Series (Yang et al., 2021), which adapts an acoustic model (AM) from speech recognition to time series classification by editing a time series into a format suitable for the AM. Recently, Chang et al. (2023) proposes LLM4TS for time series forecasting using LLMs. It designs a two-stage fine-tuning process on the LLM - first supervised pre-training on time series, then task-specific fine-tuning. Zhou et al. (2023a) leverages pre-trained language models without altering their self-attention and feedforward layers. This model is fine-tuned and evaluated on various time series analysis tasks and demonstrates comparable or state-of-the-art performance by transferring knowledge from natural language pre-training. Distinct from these approach, we neither edit the input time series directly nor fine-tune the backbone LLM. Instead, as illustrated in Fig. 1(c), we propose reprogramming time series with the source data modality along with prompting to unleash the potential of LLMs as effective time series machines.

3 METHODOLOGY

Our model architecture is depicted in Fig. 2. We focus on reprogramming an embedding-visible language foundation model, such as Llama (Touvron et al., 2023) and GPT-2 (Radford et al., 2019), for general time series forecasting *without* requiring any fine-tuning of the backbone model. Specifically, we consider the following problem: given a sequence of historical observations $\mathbf{X} \in \mathbb{R}^{N \times T}$ consisting of N different 1-dimensional variables across T time steps, we aim to reprogram a large language model $f(\cdot)$ to understand the input time series and accurately forecast the readings at H future time steps, denoted by $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times H}$, with the overall objective to minimize the mean square errors between the ground truths $\mathbf{Y}$ and predictions, i.e., $\frac{1}{H} \sum_{h=1}^H \|\hat{\mathbf{Y}}_h - \mathbf{Y}_h\|_F^2$.

Our method encompasses three main components: (1) input transformation, (2) a pre-trained and frozen LLM, and (3) output projection. Initially, a multivariate time series is partitioned into N